S.E.D.O.S.S.

How to run and test the project

Farmer hub website, virtual Raspberry Pi and the AI models. PCCOE International Grand Challenge 2026.

This guide takes you from nothing to a running farmer hub in your browser, then through a checklist to try every part of it. Allow about 20 minutes the first time. Copy each command exactly; every grey box is one command to paste into a terminal and run.

What you will run

Part	What it does
AI models	XGBoost models trained per orchard site on 10 years of NASA weather. They forecast crop water use and soil drying, and rules turn that into the day's irrigation and fertilizer decision.
Virtual Pi	A simulation of our prototype bench from the parts budget: the Raspberry Pi 5 runs the AI each morning, two ESP32s run the pumps, desalination (FDI) and nutrient dosing.
Farmer hub	The website a durian grower uses: today's decision, live unit readings, controls, a map of trained sites (and adding new ones), history and alerts.

Everything about the unit itself (tanks, FDI cell, dosing, soil) is **simulated** with placeholder numbers, and the hub says so. The AI decisions use the real trained models.

1. Before you start

You need	How to check
Python 3.10 or newer	`python3 --version` (Windows: `python --version`)
Git	`git --version`
Internet	Needed for installing packages and downloading NASA weather the first time
Chrome, Edge or Safari	To open the hub

Windows users: wherever this guide says `python3`, type `python` instead.

2. Get the code

Open a terminal (Mac: Terminal app; Windows: PowerShell) and run these one at a time:

```
git clone https://github.com/dylanchong0820/PCCOE-IGC.git
```

```
cd PCCOE-IGC
```

```
git checkout feature/farmer-hub
```

The last command picks the branch with the hub. Once the pull request is merged into main, you can skip it.

3. Install (one time only)

Create a private Python environment for the project so it does not clash with anything else:

```
python3 -m venv .venv
```

Turn it on. Do this every time you open a new terminal for the project:

Computer	Command
Mac / Linux	`source .venv/bin/activate`
Windows (PowerShell)	`.venv\Scripts\Activate.ps1`

Your prompt now starts with `(.venv)`. Then install the packages:

```
pip install -r ml/requirements.txt
```

Mac only: XGBoost also needs the OpenMP library. If you have Homebrew, run:

```
brew install libomp
```

4. Test the AI models

Move into the ml folder:

```
cd ml
```

4a. Ask for today's decision at a trained site

```
python3 predict_today.py --site raub --history examples/recent_weather_example.csv --readings examples/readings_example.json
```

You should see: a recommendation for Raub with crop water demand, the soil water change, and a DECISION line (with this example: irrigate 15 mm in the evening because of a heatwave).

4b. Ask using map coordinates instead of a site name

```
python3 predict_today.py --lat 3.5 --lon 101.9 --history examples/recent_weather_example.csv --readings examples/readings_example.json
```

You should see: "Trained site used: Raub, Pahang (32.6 km away)". Try `--lat 6.0 --lon 116.1` (Sabah) and it warns that the orchard is far from the nearest site.

4c. Retrain one site (optional, about 15 seconds)

```
python3 train_sites.py --only raub
```

You should see: it downloads Raub's NASA weather, trains, and ends with "Wrote .../sites/index.json with 6 trained site(s)".

5. Test the virtual Pi (no website)

Still in the ml folder, run a 60-day simulation of the prototype bench. The first run downloads that site's weather, so allow a few extra seconds.

```
python3 run_bench.py
```

You should see: a summary of watering days, litres delivered, FDI electrode cleanings, grams of salt recovered and how well the nutrient dosing held its target. The chart is saved as `ml/sedoss_outputs/bench_sim/fig_bench.png`.

Then go back to the project folder:

```
cd ..
```

6. Run the farmer hub

```
python3 ml/run_bench.py --serve
```

You should see: "Farmer hub on http://127.0.0.1:8765/".

Open **http://127.0.0.1:8765** in your browser. **Keep this terminal open**: the website and the simulated unit only run while this command is running. To stop it, click the terminal and press **Ctrl+C**.

7. Try every part of the hub

Work down this list with the hub open. Tick each one off.

	Do this	You should see
1	Stay on **Today**	Before 05:00: "Decision at 05:00". After that: Do / Avoid lists and a "Why" line in plain words
2	Set **Simulation speed** (left bar) to "1 hour per second"	The clock races; at midnight the calendar page tears off and the next date appears
3	**Irrigate now**, enter 5 mm, tick **Add nutrients**, **Start watering**	A progress bar under the date and the irrigation water's salt level rising to about 0.78
4	**Skip today**, then **Confirm**	The page says "Skipped by you"
5	**Pause unit**, then **Resume unit**	"Unit paused", then back to normal
6	Open **Unit**	The water path from raw tank to test bed, tanks, sensors and the parts list. Try **Clean FDI electrodes now**
7	Open **Sites** and type "Raub" in the search	The list filters to Raub
8	Click the map near Kuantan	"Nearest trained site: Segamat, 219 km away" with a red warning
9	Click **Use** on another site	The hub returns to Today; the left bar shows the new site
10	**Add a new site**: name "Kota Bharu, Kelantan", latitude 6.13, longitude 102.24, **Add and train**	A progress line for up to a minute, then "is ready"; the site appears on the map and in the list
11	Try **Add and train** with the name empty	"Give the site a name."
12	Wait a few simulated days, open **History**	A soil-water chart and one row per day
13	Open **Alerts**, click **Needs attention**	Warnings only, such as the nutrient safety cap
14	Press Ctrl+C in the terminal	Within seconds the hub shows "No signal". Start the command again and it recovers

Step 10 adds a real site to your copy of the project (new files under `ml/config/sites/` and `ml/sedoss_outputs/sites/`). Do not commit them unless the team wants that site.

8. Things that look odd but are expected

What you see	Why
Dates are in 2019	The unit replays real 2019 NASA weather so the AI works on known conditions.
"Safety cap" warnings	The AI's full nutrient dose is above the crop-safe salt limit (0.5 ppt), so the unit holds it at 0.78 dS/m. The fertilizer setting needs team review.
Irrigation water briefly red while watering	The dosing overshoots for a few seconds, then settles.
Map background blank	Map tiles need internet. Sites, markers and everything else still work.

9. If something goes wrong

Problem	Fix
Browser: "This site can't be reached"	The hub is not running. Start `python3 ml/run_bench.py --serve` and keep that terminal open.
`ModuleNotFoundError`	The environment is off or packages are missing. Run the activate command from step 3, then `pip install -r ml/requirements.txt`.
XGBoost error mentioning `libomp` (Mac)	`brew install libomp`
"Address already in use"	Another copy is already running, or the port is busy. Close the other terminal, or run with `--port 8766` and open http://127.0.0.1:8766.
"NASA weather download failed" or synthetic-data warning	Check the internet connection and run again. Sites trained on fake weather are never listed.
`python3` not found (Windows)	Use `python`, or install Python from python.org and tick "Add to PATH".
`index.json not found`	Run `python3 ml/train_sites.py` once (about 2 minutes for all sites).

10. Where things live

Path	What it is
`web/`	The farmer hub website (HTML, CSS, JavaScript)
`ml/run_bench.py`	Virtual Pi: batch simulation, or `--serve` for the hub
`ml/virtual_pi/`	Bench physics, ESP32 and Pi control code, the hub's server
`ml/config/bench.yaml`	Bench settings and what each budget part does (placeholders marked)
`ml/config/sites/`	One file per trained orchard site
`ml/train_sites.py`	Trains every site and writes the site list
`ml/predict_today.py`	The daily decision on its own (what the real Pi runs at 05:00)
`PRODUCT.md`, `DESIGN.md`	Product facts and the hub's design rules